This manuscript recovers the six principal constructions and exhaustive rank classification in Hartley–Leemans, *On the thin regular geometries of rank four for the Janko group J1* (2005), and the halving theorem and principal generalized-cube applications in Montero–Ivić Weiss, *Locally Spherical Hypertopes from Generalised Cubes*, arXiv:2001.11082v1 (29 January 2020). The common operation replaces an endpoint reflection by the conjugate of its neighbour. Its algebraic obstruction is an overlap and a least-support condition; its geometric realization is a graph double cover. Odd-endpoint constructions and index-two halving are two cases of this same operation.

The replacement is conditional on explicitly retained background. The complete finite-group subgroup catalogue for J1, the Danzer generalized-cube construction, the classical finite/affine/compact-hyperbolic Coxeter classifications, standard convex-polytope orders, and specified toroidal lattice identifications are not proved here. The finite J1 construction certificate, including its seed search and presentation calculation, and the independent exhaustive geometry census are included in Appendices A and B and are part of the proof cost. The original J1 geometry classification is no longer a retained premise. No numerical compression factor or priority claim is assigned to this draft.

The different operation in Bonheure–Buekenhout–Leemans, *On the Petrials of thin rank 3 geometries* (2001), is deliberately excluded. It replaces faces by unlabelled vertex-edge sets traced by a chamber walk. Its natural action can acquire a kernel; recovering its main obstruction requires a separate thin-residue argument. A shared reference to a group action would not make that argument a corollary of the endpoint theorem below. In particular, its unconditional automorphism embedding is not imported: for the complete tripartite geometry with two objects per type, the original group is C2³ whereas its geometric Petrial has one face and group C2². This exclusion prevents counting an unrelated proof as a third compressed manuscript.

1. The marked torsor and the geometric recognition problem

A marking of a group G is an ordered family of nonidentity involutions r_i generating G, indexed by a finite set I. For J⊆I write G_J=⟨r_j:j∈J⟩. The **intersection property**, abbreviated IP, is

$$G_J \cap G_K = G_{J\cap K} \qquad (J, K \subseteq I). \tag{1}$$

A marking satisfying (1) is a C-group marking. Its maximal parabolics are P_i=G_(I{i}). Right cosets P_i g are objects of type i, with incidence given by nonempty intersection. The common-point property for every pairwise incident family of differently typed cosets will be called **typed Helly**. IP and typed Helly are separate requirements.

Recognition lemma 1.1. If a marking has IP and its maximal-parabolic cosets have typed Helly, their incidence system is a thin, residually connected geometry on which G acts regularly; G is its full type-preserving automorphism group.

Proof. A common coset point extends every flag to the chamber consisting of its cosets in all types. Translation takes every flag to its base flag. By IP, the stabilizer of a base flag of types S is G_(I\S), the full chamber stabilizer is trivial, and a panel stabilizer is a group ⟨r_i⟩ of order two. Thus panels have exactly two chambers. In a residue of rank at least two, its stabilizer G_(I\S) is generated by its proper maximal parabolics: every remaining marker belongs to one of them. The word expressing an element in their union gives a walk in the residue incidence graph. Hence every such residue is connected. The chamber graph is connected because the markers generate G. A type-preserving automorphism fixing one chamber fixes its unique neighbour across each panel, and consequently every chamber. Thus the full automorphism group acts freely on chambers, and the already regular G-action is the full group. ■

There is a convenient finite certificate for the second requirement. Put

$$X_{ij} = \{P_i x : x \in P_j\}, \qquad P_J = \bigcap_{j\in J} P_j.$$

Recognition lemma 1.2. For a finite group, typed Helly is equivalent to

$$\left|\bigcap_{j\in J} X_{ij}\right| = [P_J : P_J \cap P_i] \quad (i \notin J,\ |J| \geq 2). \tag{2}$$

Proof. The cosets represented in P_J form a subset of the intersection on the left, and their number is the right side. Equality of sizes is therefore equality of sets. Induct on the size of a flag. Normalize all but its last object to base cosets using the common point supplied inductively. The last coset lies in the left side of (2), so has a representative in the common subgroup of those base cosets. This is a common point for the entire flag. Singletons and pairs start the induction. Conversely, typed Helly applied to such an anchored family gives the set equality and hence (2). ■

These lemmas apply to the actual orbit partitions of the regular G-torsor. Replacing each g by its permutation matrix loses no information, provided the distinguished parabolic partitions are retained. Linearity alone proves neither (1) nor (2).

2. A single endpoint theorem

Let (G;a,b,(r_i)_(i∈E)) be a C-group marking, with [a,r_i]=1 for all i∈E. No string diagram, finite group, or underlying geometry is assumed. Define

$$a' = aba, \quad R_J = \langle r_j : j \in J \rangle, \quad U_J = \langle a, b, R_J \rangle, \quad H_J = \langle a', b, R_J \rangle,$$
$$P = \langle b, R_E \rangle, \quad H = H_E, \quad \mathcal{F} = \{J \subseteq E : a \in H_J\}. \tag{3}$$

Conjugation by a exchanges b and a′ and fixes R_J pointwise. Consequently

$$H_J \trianglelefteq U_J = H_J\langle a \rangle, \qquad [U_J : H_J] \leq 2. \tag{4}$$

Endpoint theorem 2.1. The new marking (H;a′,b,(r_i)) has IP if and only if

$$P \cap P^a = R_E, \qquad \mathcal{F} \text{ is closed under intersections.} \tag{5}$$

The group H is normal of index at most two in G. It is G precisely when $\mathcal{F}$ is nonempty. Since E is finite, a nonempty intersection-closed $\mathcal{F}$ is precisely the family of supersets of one least support S⊆E.

Proof. The family $\mathcal{F}$ is upward closed. Any new marked subgroup is of one of the forms

$$A_{\epsilon,J} = \langle b^\epsilon, R_J \rangle \ (\epsilon = 0, 1), \qquad H_J, \qquad Q_J^a \text{ with } Q_J = \langle b, R_J \rangle, \tag{6}$$

where b^0 means that b is omitted. These correspond to omitting a′, using both endpoint labels, or using a′ but omitting b. Intersections of two A's are old marked intersections. Also

$$A_{\epsilon,I} \cap H_J \leq A_{\epsilon,I} \cap U_J = A_{\epsilon,I\cap J};$$

all generators on the right belong to H_J, so equality holds. Intersections of two Q^a's follow by conjugating old IP. Conjugation by a normalizes H_I, so an H-with-Q^a intersection reduces to the just proved H-with-A case. For an A-with-Q^a intersection, the first condition in (5) confines it to R_E. Intersecting there and using that a centralizes R_E gives R_(I∩J).

It remains to intersect H_I and H_J. Put K=I∩J. Old IP gives

$$H_K \leq H_I \cap H_J \leq U_I \cap U_J = U_K. \tag{7}$$

If a∈H_K then H_K=U_K. Otherwise H_K has index two in U_K, so the middle subgroup is either H_K or U_K. The latter case occurs exactly when a belongs to both H_I and H_J, and intersection closure excludes it. This proves all new marked intersections. Necessity of the first condition follows by intersecting the two endpoint-deleted new parabolics; necessity of the second follows by intersecting H_I,H_J. The generators are involutions and the proved

IP supplies independence. Equation (4) with J=E gives the group and index assertions. A finite nonempty upward intersection-closed family contains its total intersection S, giving the least-support description. ■

Corollary 2.2 (odd endpoint and proper halving). If m=|ab| is odd, the only test in (5) is the overlap P∩P^a=R_E, and H=G. If H is proper in G, the only test is again the overlap, and [G:H]=2. In either case the endpoint product has order m/gcd(m,2) for finite m; infinite m remains infinite.

Indeed a′b=(ab)², and for odd m,

$$a = (a'b)^{(m+1)/2}b \in H_{\emptyset}.$$

Thus $\mathscr{F}$ is respectively all subsets or empty. An even endpoint order by itself does not imply that H is proper; the least-support alternative in Theorem 2.1 cannot be discarded.

Lemma 2.3 (central twist). Suppose b,c are involutions with |bc|=m odd and m≥3, and w is a nonidentity involution centralizing b,c. Replacing b by wb preserves the generated group and changes |bc| to 2m, provided w belonged to the original group.

Proof. The commuting factors w and bc have coprime orders, so (wbc)^m=w and |wbc|=2m. Hence the new generators recover w and b. The new marker wb is a nonidentity involution: wb=1 would make b=w commute with c, contrary to m≥3 odd. ■

A parabolic containing w,b,c is unchanged by this twist; one omitting b is unchanged. Other intersections and typed Helly still require proof. Lemma 2.3 is the auxiliary operation needed in the finite application below, not a blanket geometric preservation theorem.

3. The geometric meaning of the support condition

Let Γ be a finite-rank residually connected incidence geometry, of rank n≥3. Designate types 0 and 1 as vertices and edges. Assume their truncation is a simple graph Q, and that an edge is incident with a higher object exactly when both its endpoints are incident with that object. For a flag F of higher objects, let Q_F be its common vertex-edge shadow. All Q_F are connected and contain an edge.

Here is the connectivity fact being used. The chamber graph of a residually connected geometry is connected: induct on rank inside object residues, and link objects using paths in the incidence graph. Restricting such a chamber gallery to any set of at least two types shows that every corresponding truncation is connected, because successive complete incidence subgraphs overlap. Apply this also inside the residue at F; chamber extension supplies an edge. The edge-shadow hypothesis says Q_F is the induced graph on its common vertices.

For a connected graph Q, its bipartite double cover D(Q) has vertices (v,ε), ε∈{0,1}, with (v,ε) adjacent to (w,1−ε) when v is adjacent to w. It is connected when Q has an odd cycle, and has two components otherwise: parity of walks proves both assertions. Choose a component C of D(Q). Define L_C(Γ) as follows. Its first two types are (v,0) and (v,1) in C. A higher object is a pair (x,C_x), where C_x is a component of D(Q_x) contained in C. Incidence of the first two types is adjacency, incidence with a higher object is membership in C_x, and two higher objects are incident when their old objects are incident and their chosen components meet.

The precise compatibility requirement is

$$\text{if every } Q_x \ (x \in F) \text{ is nonbipartite, then } Q_F \text{ is nonbipartite,} \tag{PC}$$

for every nonempty higher flag F. It is automatic if the higher shadows of every flag are nested, or if Q is bipartite.

Geometric theorem 3.1. The system L_C(Γ) is always a geometry. It is residually connected exactly when (PC) holds. It is thin exactly when Γ is thin. A chamber-transitive automorphism group of Γ acts faithfully and chamber-transitively on L_C(Γ) when Q is nonbipartite; when Q is bipartite its index-two color-preserving subgroup does so.

Proof. For a flag Φ={(x,C_x):x∈F} of higher lifts, write W=∩C_x. If any Q_x is bipartite, restriction of C_x to D(Q_F) is one component. Any two such restrictions agree: incidence makes them meet over the connected Q_(x,y),

and hence select the same parity class there and over Q_F. Nonbipartite constituents impose no restriction. Thus W is one component if any constituent is bipartite, and all of D(Q_F) otherwise. In particular W always contains an edge, and is connected exactly under (PC) for this flag. For F empty use W=C.

An edge of W and a chamber of Γ extending its base edge and F give a chamber extending Φ. Every additional higher object has a unique component containing the chosen lift of that edge. The same extension works for a flag already containing one lifted vertex. If it contains both, their base edge is unique by simplicity and incident with all the old higher objects by the shadow condition. This proves the geometry axiom.

The residue at (v,0) is the old residue Γ_v: a neighbour (w,1) corresponds to the unique edge {v,w}, and the component for an incident higher object is uniquely specified by (v,0). Simplicity and the shadow condition preserve all incidences. The other color is identical. Hence residues of flags containing a lifted vertex are connected. For a flag of higher objects only, its two-type truncation is W; when W is connected, every other residual object attaches to it through a chamber. This proves residual connectivity under (PC). If (PC) fails, choose its higher flag; all constituents have unique lifts but W has two components. Were the resulting geometry residually connected, the truncation lemma inside its residue would make W connected. This proves necessity without treating a truncation as a residue.

Panels containing both new vertex types correspond to old panels missing the same higher type. A panel missing one new vertex type corresponds to an old panel missing the edge type, at the retained endpoint. Old panels missing the vertex type already have two objects by simplicity. These correspondences prove both directions of thinness.

If Q is nonbipartite, an old chamber (v,{v,w},x_2,…,x_(n−1)) corresponds bijectively to the new chamber with (v,0),(w,1) and the components containing them. Old automorphisms act naturally on this bijection. The action is faithful: fixing new objects fixes old vertices and higher objects, and simplicity then fixes every edge. If Q is bipartite, the same bijection uses the old chambers whose chosen vertex is in one fixed color class. A chamber-transitive group contains a color-exchanging element, since it exchanges the endpoints of a base edge through a chamber move. Its color-preserving subgroup therefore has index two and is transitive on the selected chambers, with the same faithfulness proof. ■

Suppose now Γ is regular and thin, with vertex-changing marker a, edge-changing marker b, and higher markers r_i; assume a commutes with the higher markers. Under (PC), Theorem 3.1 produces a regular thin residually connected geometry. Directly on a base edge, its panel involutions are b^a,b,r_i: b changes one endpoint, aba changes the other, and each r_i changes its higher object. Lemma 1.1's chamber argument identifies their group as the full type-preserving group H. Conversely, walking within residues identifies the stabilizer of a base flag with the subgroup generated by its missing-type markers, giving IP and identifying the actual coset incidence with L_C(Γ).

There is also an exact correspondence between the two parity conditions. In the base residue with group U_J from (3), the vertex stabilizer is Q_J=⟨b,R_J⟩. Its vertex graph is bipartite precisely when a∉H_J. If H_J is proper, its two cosets give a coloring, since Q_J≤H_J and a crosses colors. Conversely the action on the two colors supplies a character killing b and all r_j and sending a to the nontrivial element, so it excludes a from H_J. In a globally nonbipartite geometry, higher flags reduce by transitivity to base flags with shadows U_(E\S). Condition (PC) says

$$[E \setminus \{i\} \in \mathcal{F} \text{ for all } i \in S] \Longrightarrow E \setminus S \in \mathcal{F}. \tag{8}$$

An upward family containing E satisfies (8) exactly when it is principal: let T be the set of i whose complementary coatom belongs to $\mathcal{F}$; (8) gives E\T∈$\mathcal{F}$, and every member of $\mathcal{F}$ contains E\T. Thus geometric parity compatibility is the least-support condition of Theorem 2.1. The mixed overlap in that theorem has a different role: it says the two adjacent vertices determine one edge. Indeed P and P^a stabilize the endpoints; their intersection is the edge's pointwise stabilizer R_E exactly when no other edge has those endpoints, using transitivity on panels and the regular chamber identification.

4. All lattice-polytope halvings

Corollary 4.1. Let P be a regular rank-n abstract polytope, n≥3, whose face poset is a lattice. Replacing one endpoint marker a by aba gives a C-group H normal of index at most two in Aut(P). Its maximal-parabolic coset system is a regular hypertope, with full type-preserving group H. The diagram duplicates the edge labels from b to the remaining

markers, and the a′–b label is m/gcd(m,2) for m=|ab| finite, or infinity for infinite m. The index is two exactly when the corresponding vertex graph is bipartite; for a final-end operation use the dual graph on facets.

Proof. In a lattice polytope an edge is the join of its two distinct vertices. Thus two vertices determine at most one edge, and that edge belongs to a higher face exactly when both vertices do. Higher faces of a flag form a chain, so their shadows are nested and (PC) is automatic. Theorem 3.1 and the panel-marker identification prove the asserted geometry, full group, and IP without an alternating-semiregular-polytope theorem. Equation (4) and the coloring argument give the index. Finally a′b=(ab)² and a commutes with all r_i, so a′r_i is conjugate to br_i; all other products are unchanged. ■

This is N013 Theorem 3.1, Proposition 3.2 and Corollary 3.3, after reversing their final-end convention. Their term *non-degenerate* is used in the lattice-face-poset sense, as explicitly explained by the later primary halving manuscript cited in the references; a general abstract polytope with simple diagram is not substituted for that hypothesis. N013's auxiliary direct-product Lemma 2.4 follows at once because a marked subgroup of a product with disjoint factor markings is the product of the corresponding marked subgroups, and intersections are coordinatewise. Its quoted Propositions 2.1 and 2.3 are background; the latter is not needed. Recognition lemma 1.1 supplies the content needed from its quoted Theorem 2.2.

5. Generalized cubes: one parity module gives every principal application

Retain the following Danzer construction. If K is a finite regular lattice polytope with m vertices, 2^K has binary vertex set V=F_2^m. A face associated with a face F of K and x∈V allows exactly the coordinates indexed by the vertices of F to vary. These coordinate subcubes, ordered by inclusion with their improper faces, form a regular lattice polytope of one higher rank, with vertex figures K and automorphism group

$$\mathrm{Aut}(2^K) = V \rtimes \mathrm{Aut}(K). \tag{9}$$

The coordinate action is the natural vertex permutation action. This construction theorem is a substantial retained input; neither its polytopality nor the identification of the full group in (9) is inferred solely from an order calculation.

Corollary 5.1. Halving 2^K at its vertex endpoint gives the exact group

$$E_m \rtimes \mathrm{Aut}(K), \qquad E_m = \{x \in \mathbf{F}_2^m : \sum x_i = 0\}, \tag{10}$$

of order 2^(m−1)|Aut(K)|, acting as the full type-preserving group of the regular hypertope in Corollary 4.1.

Proof. The vertex graph of 2^K is the m-cube, bipartitioned by coordinate parity. Its color-preserving subgroup of (9) is exactly (10), so Corollary 4.1 identifies the whole halving group. Alternatively, let a flip the coordinate of a base vertex v of K and let b be its first distinguished reflection, carrying v to an adjacent vertex v′. The product a′b=(aba)b is translation by e_v+e_v′. The unchanged reflections generate Aut(K). Their conjugates give every edge difference in the connected vertex graph of K; along paths, these span E_m. This independently proves generation of the displayed subgroup. ■

N013 uses the dual convention P=(2^(K*))*. Consequently m in its formulas is the number of facets of K, and reversing all types converts (10) exactly to its construction.

Corollary 5.2 (polygon inputs). For a p-gon K, p≥3, the result is a self-dual orientable regular map of type {p,p}, with

$$V = F = 2^{p-1}, \qquad E = p2^{p-2}, \qquad |\,\mathrm{Aut}\,| = p2^p, \qquad g = 1 + 2^{p-3}(p-4). \tag{11}$$

Proof. The two vertex colors give V and F after ordering the map types as (color 0, old square, color 1). Each edge of the p-gon specifies 2^(p−2) coordinate squares, giving E. The fork diagram becomes {p,p}. A single coordinate flip conjugates the two new endpoint reflections and exchanges the colors, supplying the duality. Extend the reflection-sign character of the dihedral Aut(K) trivially over E_m. Each of the three new distinguished reflections has sign −1, so

the chamber orientation is well defined and reverses across every adjacency. This proves orientability. Euler's formula V−E+F=2−2g gives (11). ■

For p=3 the map is the tetrahedron: the group is $E_3 \rtimes S_3$ of order 24 with the spherical A_3 marking and its universal Coxeter geometry. For p=4 it is the square toroid conventionally denoted {4,4}_(2,2). Identifying that particular translation lattice, and identifying the original dual generalized cube with {4,4}_(4,0), are retained conventional grid identifications, not deductions from equal orders.

Corollary 5.3 (all locally spherical inputs and exceptional orders). In output rank n≥4, the locally spherical cases of N013's dual generalized-cube construction are obtained from K equal to the (n−1)-simplex or (n−1)-cube, and additionally the dodecahedron for n=4 and the 120-cell for n=5. Their types are respectively spherical D_n, affine B_(n−1), and the rank-four/rank-five compact-hyperbolic forks. The last two finite groups have orders

$$2^{11} \cdot 120 = 245760, \qquad 2^{119} \cdot 14400. \tag{12}$$

Proof. Put L=K* to use the coordinate model. Each of its two new vertex residues is L itself. If the output is locally spherical, L must therefore be a universal spherical polytope. Retain the classical finite string-Coxeter list. In rank at least four a proper three-node fork with equal labels q,q is spherical only for q=3, where q is the first label of L. In rank five, the remaining candidate F_4={3,4,3} is excluded because its proper four-node fork is affine. The finite list leaves exactly the simplex, cross-polytope, icosahedron and 600-cell, the last two only in their respective ranks. For rank at least six only the first two remain.

Conversely, all proper faces of these L are simplices. At an old j-face of 2^L, j≥2, let F be its associated (j−1)-face of L. The lower and upper sections vary independently. After halving, the maximal residue is the product of the compressed cube 2^F with the spherical section L/F. As F has j vertices, the first factor has group E_j⋊S_j and its D_j marking; its order 2^(j−1)j! equals that of the finite Coxeter group D_j, proving equality with the universal spherical geometry. Here $D_2=A_1\times A_1$ and $D_3=A_3$. Vertex residues are L, so all maximal residues, and consequently all proper residues, are genuinely spherical geometries. A diagram alone is not used as a substitute for those residues.

The new global diagrams follow from Corollary 4.1. The retained finite, affine and compact-hyperbolic Coxeter classifications identify them as stated. Formula (10), with m=12 or 120 and the classical orders |Aut(dodecahedron)|=120 and |Aut(120-cell)|=14400, gives (12). For the simplex m=n and |Aut(K)|=n!, giving D_n order 2^(n−1)n!. For the (n−1)-cube, m=2(n−1) and |Aut(K)|=2^(n−1)(n−1)!, giving 2^(3n−4)(n−1)!. The finite hyperbolic examples are proper hypertopes because the fork diagrams have a vertex of degree three. ■

The source's toroidal notation for cube input is {4,3^(n−4),3/3}_(4,0,…,0); both the original cubical toroid {4,3^(n−3),4}_(4,0,…,0) and the halved lattice label are retained standard affine-lattice identifications. They are separate from the exact group, order and residue proofs just given. N013's claim that its rank-four hyperbolic example differs from every example in its reference [9] is not used or independently verified. Its auxiliary identification with an alternating semiregular polytope is likewise unnecessary and retained as background rather than counted as a new corollary.

There is a scope correction to N013 Remark 4.1. Retaining Pellicer's extension construction, the halved diagram for endpoint 2s has a three-node fork triangle with labels (p,p,s). For output rank at least four and p,s≥3, this proper rank-three diagram is not finite, since 2/p+1/s≤1; thus the stated non-local-sphericity follows in those ranks. In rank three all proper residues are finite dihedral for finite parameters, so the blanket non-local-sphericity wording is not valid under the source's own definition. Neither this remark nor its correction is a new open-problem claim.

6. Six finite constructions from one marking

The second application takes the odd branch of Corollary 2.2. Retain the GAP TomLib J1 permutation model, of order 175560. Appendix A constructs a marking (a,b,c,d) within that model with

$$\begin{aligned} |ab|, |ac|, |ad|, |bc|, |bd|, |cd| &= (5, 2, 2, 3, 2, 5), \\ (abc)^5 = ((dcb)^5 a)^3 &= 1. \end{aligned} \tag{13}$$

It also completes a Todd–Coxeter calculation for the literal presentation with the four involution relations and (13): the index over ⟨b,c,d⟩ is 1463. That subgroup is a quotient of the finite Coxeter group [3,5] of order 120, so the presented group has order at most 1463·120=175560. The found generating tuple in the TomLib model supplies a quotient of that order; hence this is the source presentation of J1. A valid permutation action alone would not prove the upper bound: the completed exact coset enumeration is a charged computational input.

Set w=(bcd)^5. The certificate proves that w is a nonidentity central involution of ⟨b,c,d⟩. The six source-defined markings and their exact invariants are:

Geometry	Marking	Pair orders ab,ac,ad,bc,bd,cd in that marking	Maximal-parabolic orders
Γ1	(a,b,c,d)	5,2,2,3,2,5	120,20,20,60
Γ2	(a,wb,c,d)	5,2,2,6,2,5	120,20,20,660
Γ3	(a,b,c,c^d)	5,2,2,3,3,5	120,20,60,60
Γ4	(b^a,b,c,d)	5,3,2,3,2,5	120,120,20,60
Γ5	($b^{a,b,c,c}$d)	5,3,3,3,3,5	120,120,60,60
Γ6	((wb)^a,wb,c,d)	5,6,2,6,2,5	120,120,20,660

Order-120 parabolics are C2×A5, order-20 ones are D20=C2×D10, order-60 ones A5 and order-660 ones $PSL_2(11)$; D10 here has order ten. Structure identifications use the retained exact GAP algorithms and named group models.

Corollary 6.1. Each of these six marked coset systems is a thin regular residually connected rank-four geometry with full type-preserving group J1. They are pairwise nonisomorphic, even allowing type permutations. These are L088 Theorems 3.2, 4.2, 5.2, 5.4, 6.2 and 7.2, including their generation lemmas.

Proof. Lemma 2.3 gives the central twist Γ1→Γ2 and preserves its first parabolic. Every endpoint operation has odd endpoint order five, so Corollary 2.2 recovers the same group. Appendix A verifies the two actual subgroup equalities

$$\langle b,c,d\rangle \cap \langle b,c,d\rangle^a = \langle c,d\rangle, \qquad \langle a,b,c\rangle \cap \langle a,b,c\rangle^d = \langle a,b\rangle. \tag{14}$$

They have order ten. The second gives Γ3, the first Γ4. In Γ3, ⟨b,c,c^d⟩=⟨b,c,d⟩ because |cd|=5, and ⟨c,c^d⟩=⟨c,d⟩; hence the same first equality gives Γ5. In Γ2 its first parabolic is unchanged by the twist, so that equality gives Γ6. Thus after the seed and twist cases, four IP proofs reduce to the two meets (14). Appendix A additionally verifies all 256 ordered marked-subset meets for each of the six markings, so the finite result does not depend on a mistaken reuse of a subgroup identity.

For each marking it checks all sixteen equalities (2), without presuming that a C-group with a nonstring diagram is a geometry. Recognition lemmas 1.1–1.2 now give all the geometric conclusions. The six multisets of maximal-parabolic orders in the table are distinct, so no type-permuting isomorphism identifies two geometries. ■

The definition of Γ3 in L088 §5 duplicates an A5 parabolic, whereas its printed figure labelled Γ3 duplicates C2×A5; the definition of Γ4 has the reverse discrepancy. The source figure labels Γ3 and Γ4 are therefore transposed relative to its definitions. The table follows the definitions. This is a source correction, not a seventh construction.

The two source-mentioned failures also follow from the same certificate. The markings (a,wb,c,c^d) and (a,b,c,wd) still generate J1 and satisfy typed Helly, but each has eight failing ordered IP meets; their panel orders are respectively (2,4,2,2) and (2,2,2,4). Therefore they are not thin. These failures demonstrate why neither endpoint generation nor flag transitivity is a substitute for IP.

6.2. Exhaustiveness by a general residue-extension theorem

The original geometry census can be replaced by a short complete search on proper residues. The required group data are the complete TomLib table of J1 subgroup-conjugacy classes, not a list of its geometries.

Residue-extension theorem 6.2. Let G be finite and let $M_1,\ldots,M_s$ represent all conjugacy classes of maximal proper subgroups. For each M_j enumerate, up to M_j-conjugation, all unordered k-element sets of involutions whose markings satisfy IP and typed Helly. Append every involution of G outside the subgroup generated by each retained set and apply the same tests. This finds every rank-(k+1) thin regular residually connected geometry whose full type-preserving group is a subgroup of G. To restrict to group G, retain exactly the extensions generating G.

Proof. For any such rank-(k+1) geometry, its distinguished marking consists of independent involutions. Omitting one gives a proper subgroup P of its full group and hence of G, so P lies in a conjugate of one M_j. Its marked coset geometry is the corresponding rank-k residue, thus satisfies both IP and typed Helly. Conjugation into M_j and then within M_j puts its unordered markers in the enumerated list. Its omitted involution lies outside P and occurs in the extension loop. Conversely, the tested IP and Helly conditions give exactly the asserted geometries by Lemma 1.1. Ordering the marker set merely orders the type labels. ■

For completeness, the residue statement used here requires no extension of arbitrary residue automorphisms to G. The base-flag stabilizer acts transitively and freely on the residue's chambers, and the residue is thin and connected. Its full type-preserving group also acts freely, so the stabilizer already is that full group. Walking inside the residue identifies its panel generators. Thus local Helly is a necessary condition for these genuine residues, not an arbitrary extra filter.

Corollary 6.3. Up to isomorphism allowing a permutation of types, the six geometries in Corollary 6.1 are all the thin regular residually connected rank-four geometries with group J1. There are none of rank at least five.

Proof. Appendix B executes Theorem 6.2 for k=3 and k=4. The retained complete table has forty subgroup classes. Its nonzero containment entries determine seven maximal classes of orders 660,168,120,114,110,60,42. The local counts are:

Maximal subgroup order	Involutions	Triple orbits	Quadruple orbits	Regular triple orbits	Regular quadruple orbits
660	55	60	588	42	6
168	7	3	3	2	0
120	31	108	616	45	7
114	19	12	42	0	0
110	11	2	4	0	0
60	23	58	221	6	2
42	7	2	2	0	0
Total	—	245	1476	95	15

The conjugation permutation action on involutions need not be faithful; its image nevertheless has exactly the required orbits on subsets. No local generated subgroup is required to equal its containing maximal subgroup. The 95 retained triples reduce to 70 J1-conjugacy classes. All 100016 possible outside-involution extensions are tested; forty pass with generated group J1, collapsing to precisely six classes. Their maximal-parabolic order multisets are the six distinct ones in Corollary 6.1, which already supplies six pairwise nonisomorphic constructions.

Here is the exact conjugacy test, so equality of numerical invariants is not mistaken for isomorphism. There is one class of 1463 involutions in J1, with centralizer order120. Fix a_0. For each x in an unordered marker set T choose g_x with x^g_x=a_0, enumerate T^(g_xc) for every c∈C_G(a_0), and take the least sorted tuple of involution indices. These are exactly the conjugates of T containing a_0: every conjugator carrying x to a_0 differs from g_x on the right by such a c. Consequently keys agree exactly for inner-conjugate unordered markings. There is no assumption that all automorphisms of J1 are inner. The six different parabolic-order multisets prevent any further isomorphism from merging those classes, while the complete six-class upper bound prevents any additional geometry.

For k=4, the fifteen local regular quadruple classes remain fifteen after J1-conjugacy deduplication. All 21352 extensions by an involution outside their generated subgroup fail the full regular-geometry tests. This stage does not discard proper-subgroup targets. If a regular geometry of rank r≥5 existed for J1, fixing r−5 objects of a base chamber would yield a rank-five regular residue with group contained in J1; Theorem 6.2 would have found it. Thus all higher ranks are excluded. This does not assert nonexistence of C-group markings that fail typed Helly. ■

The source's successful central modifications in §8, when they generate J1 and yield a thin regular geometry, are consequently covered by the same exhaustive classification. The two displayed failures were checked separately above. Appendix A also verifies the incidental source equality N_G(⟨a,b,c⟩)=⟨a,b,c⟩. Its initial vertex figure is the universal spherical [3,5] geometry, since that parabolic has the full Coxeter order120; its facet is the antipodal hemi-dodecahedral quotient of [5,3], since the A5 quotient has order60 and kernel the unique central order-two subgroup of C2×A5. These conventional spherical identifications use the named classical Coxeter models.

7. Scope, reuse and proof accounting

The common theory has two genuinely reused parts: the endpoint theorem identifies the normal index-two interval and the precise mixed-intersection obstruction, and the recognition lemmas distinguish group IP from geometric flag completion. In the finite J1 application the odd endpoint collapses the support test and exact coset arithmetic supplies Helly. In lattice-polytope applications the graph double cover proves flag completion and forces the support condition through nested shadows. The generalized-cube order formulas then become a single parity-subspace calculation rather than separate group constructions.

This does not follow directly from the newer marked parabolic comparison theorem in this repository. That theorem compares a marking and its elementwise homomorphic image, assuming injectivity on products of protected parabolics and an exact exceptional-membership condition. Replacing a by aba while leaving b fixed does not provide such a homomorphism in general, and no required product-set injectivity has been established for a nontrivial common quotient here. Citing that theorem without those hypotheses would add notation, not shorten these adapters. The present proof accordingly does not import it or charge it as a proof shortcut.

The finite proof certificates are the complete GAP programs in Appendices A and B. Their separate executable files are byte-identical to those blocks; the runner checks this before executing in a fresh directory, records GAP/TomLib versions and the J1 data hash, and preserves the complete logs. Appendix A proves the six constructions and two negative controls; Appendix B independently exhausts the regular geometries through the residue-extension theorem. The algebraic and all-parameter geometric results in §§1–5 are written proofs; finite diagnostics are not substituted for them. No new Lean formalization of this complete manuscript is asserted. Earlier endpoint mixed-intersection and index-two formalizations certify narrower generic lemmas only.

For any later measurement, count the whole manuscript from its introduction through Appendix B, including this scope discussion and all code. The executable duplicate is counted once, while runner and output bytes are reported separately. Retained background remains a conditional dependency cost, especially the complete J1 subgroup catalogue and Danzer/Coxeter inputs. The source intake and statement ledger explicitly record the unproved comparative claim and rank-qualified remark. This draft does not yet pass an independent assembled-manuscript review and receives no compression factor until it does.

References and retained inputs

1. M. I. Hartley and D. Leemans, *On the thin regular geometries of rank four for the Janko group J1*, Innovations in Incidence Geometry **1** (2005), 181–190. Journal primary PDF. Main construction targets: Theorems 3.2, 4.2, 5.2, 5.4, 6.2, 7.2. Its introductory exhaustive classification is recovered in Corollary 6.3.
2. A. Montero and A. Ivić Weiss, *Locally Spherical Hypertopes from Generalised Cubes*, arXiv:2001.11082v1, 29 January 2020. Targets: Theorem 3.1, Proposition 3.2, Corollary 3.3, and the principal applications in §4. Later journal numbering is not substituted.
3. C. A. Piedade and P. Tranchida, *Constructing new geometries: a generalized approach to halving for hypertopes*, arXiv:2405.19050v1, especially its introduction and §2.3 on lattice nondegeneracy, its graph constructions, and the established bipartiteness/index-two interpretation. This is attribution and terminology support, not an additional compressed manuscript or an import of its broader geometric conclusion without (PC).
4. L. Danzer, *Regular incidence-complexes and dimensionally unbounded sequences of such. I*, North-Holland Mathematics Studies **87** (1984), 115–127; P. McMullen and E. Schulte,* Abstract Regular Polytopes* (2002), especially the generalized-cube construction and standard coordinate models. The construction theorem in (9), conventional toroidal lattice identifications and standard convex-polytope orders remain retained.
5. The classical finite/affine/compact-hyperbolic Coxeter classifications, and the GAP 4.15.1 exact permutation-group/coset algorithms with TomLib 1.2.11's complete J1 subgroup catalogue and permutation model, are the

other substantial inputs. The execution manifest binds the actual installed versions and data.

Appendix A. Complete finite construction certificate

The following code is part of this manuscript, not an excluded implementation appendix. It searches for its own source marking in the retained ambient group, rather than reading an earlier geometry census or a hard-coded witness. It then verifies the presentation upper bound, every marked intersection and all required typed-Helly equalities.

```
# Entire finite certificate for Compress_C. No geometry census is read.
LoadPackage("tomlib");;
SizeScreen([100000,24]);;
CheckC:=function(test,msg) if not test then Error(msg);fi;end;;
G:=UnderlyingGroup(TableOfMarks("J1"));;
CheckC(Size(G)=175560,"ambient J1 order");;
FindSeedC:=function(K)
 local a,b,c,d,iv,bs;
 a:=GeneratorsOfGroup(K)[1];
 CheckC(Order(a)=2,"ambient first generator");
 iv:=Filtered(Elements(Centralizer(K,a)),x->Order(x)=2);
 for d in iv do
  if d=a then continue;fi;
  bs:=Filtered(Elements(Centralizer(K,d)),x->Order(x)=2 and Order(a*x)=5);
  for c in iv do
   if Order(c*d)<>5 then continue;fi;
   for b in bs do
    if Order(b*c)=3 and (a*b*c)^5=One(K)
      and ((d*c*b)^5*a)^3=One(K) and Group(a,b,c,d)=K then
     return [a,b,c,d];
    fi;
   od;
  od;
 od;
 Error("no source seed");
end;;
s:=FindSeedC(G);;
pairs:=Combinations([1..4],2);;
CheckC(List(pairs,p->Order(s[p[1]]*s[p[2]]))=[5,2,2,3,2,5],"seed pairs");;
CheckC(Normalizer(G,Group(s{[1,2,3]}))=Group(s{[1,2,3]}),"seed facet normalizer");;
# Exact upper bound for the literal source presentation.
F:=FreeGroup("a","b","c","d");; u:=GeneratorsOfGroup(F);;
rels:=Concatenation(List(u,x->x^2),List([1..6],i->
 (u[pairs[i][1]]*u[pairs[i][2]])^[5,2,2,3,2,5][i]),
 [(u[1]*u[2]*u[3])^5,((u[4]*u[3]*u[2])^5*u[1])^3]);;
Q:=F/rels;; q:=GeneratorsOfGroup(Q);;
ct:=CosetTable(Q,Subgroup(Q,q{[2,3,4]}));;
CheckC(Length(ct[1])=1463,"presentation coset index");;
CheckC(Size(Group(List([1,3,5,7],i->PermList(ct[i]))))=175560,"presentation image");;
w:=(s[2]*s[3]*s[4])^5;;
CheckC(Order(w)=2 and ForAll(s{[2,3,4]},x->Comm(w,x)=()),"twist center");;
marks:=[s,[s[1],w*s[2],s[3],s[4]],
 [s[1],s[2],s[3],s[3]^s[4]],
 [s[2]^s[1],s[2],s[3],s[4]],
 [s[2]^s[1],s[2],s[3],s[3]^s[4]],
 [(w*s[2])^s[1],w*s[2],s[3],s[4]]];;
CheckC(Intersection(Group(s{[2,3,4]}),Group(s{[2,3,4]})^s[1])
```

```
 =Group(s{[3,4]}),"left overlap");;
CheckC(Intersection(Group(s{[1,2,3]}),Group(s{[1,2,3]})^s[4])
 =Group(s{[1,2]}),"right overlap");;
VerifyC:=function(t)
 local ss,subs,a,b,ip,pars,i,j,J,rows,T,common,expected,h,pan;
 ss:=Combinations([1..4]);
 subs:=List(ss,J->Group(Concatenation([()],t{J})));
 ip:=0;
 for a in [1..16] do for b in [1..16] do
  if Intersection(subs[a],subs[b])<>subs[Position(ss,Intersection(ss[a],ss[b]))]
   then ip:=ip+1;fi;
 od;od;
 pars:=List([1..4],i->Group(t{Difference([1..4],[i])}));
 h:=[];
 for i in [1..4] do
  rows:=[];
  for j in Difference([1..4],[i]) do
   T:=Intersection(pars[i],pars[j]);
   rows[j]:=Set(List(RightTransversal(pars[j],T),x->RightCoset(pars[i],x)));
  od;
  for J in Combinations(Difference([1..4],[i])) do
   if Length(J)<2 then continue;fi;
   common:=Length(Intersection(rows{J})); T:=Intersection(pars{J});
   expected:=Index(T,Intersection(T,pars[i]));
   Add(h,[i,J,common,expected]);
  od;
 od;
 pan:=List([1..4],i->Size(Intersection(pars{Difference([1..4],[i])})));
 return rec(order:=Size(Group(t)),pairs:=List(pairs,p->Order(t[p[1]]*t[p[2]])),
  parabolics:=List(pars,Size),types:=List(pars,StructureDescription),
  ip_failures:=ip,helly:=h,borel:=Size(Intersection(pars)),panels:=pan);
end;;
expectedP:=[[120,20,20,60],[120,20,20,660],[120,20,60,60],
 [120,120,20,60],[120,120,60,60],[120,120,20,660]];;
expectedPairs:=[[5,2,2,3,2,5],[5,2,2,6,2,5],[5,2,2,3,3,5],
 [5,3,2,3,2,5],[5,3,3,3,3,5],[5,6,2,6,2,5]];;
for i in [1..6] do
 CheckC(ForAll(marks[i],x->Order(x)=2),"marker orders");
 r:=VerifyC(marks[i]);
 CheckC(r.order=175560 and r.ip_failures=0,"generation and IP");
 CheckC(r.parabolics=expectedP[i] and r.pairs=expectedPairs[i],"source tables");
 CheckC(r.types=List(r.parabolics,o->["C2 x A5","D20","A5","PSL(2,11)"]
  [Position([120,20,60,660],o)]),"parabolic isomorphism types");
 CheckC(r.borel=1 and r.panels=[2,2,2,2],"thinness");
 CheckC(ForAll(r.helly,z->z[3]=z[4]),"typed Helly");
 Print("GEOMETRY ",i," PAIRS ",r.pairs," PARABOLICS ",r.parabolics,
  " TYPES ",r.types," IP_MEETS 256 HELLY_ROWS ",Length(r.helly)," PASS\n");
od;
CheckC(Length(Set(List(expectedP,SortedList)))=6,"pairwise nonisomorphism");;
bad:=[[s[1],w*s[2],s[3],s[3]^s[4]],[s[1],s[2],s[3],w*s[4]]];;
for i in [1..2] do
 r:=VerifyC(bad[i]);
 CheckC(r.order=175560 and r.ip_failures=8 and r.borel=1,"negative scope");
 CheckC(r.panels=[[2,4,2,2],[2,2,2,4]][i],"negative panel");
 CheckC(ForAll(r.helly,z->z[3]=z[4]),"negative Helly");
```

```
 Print("NEGATIVE ",i," IP_FAILURES ",r.ip_failures," PANELS ",r.panels," PASS\n");
od;
Print("COMPRESS_C_J1_PASS\n");
QUIT;
```

Appendix B. Complete exhaustive residue-extension certificate

This program reads no geometry representative list. It derives the maximal-subgroup classes from the complete group table, enumerates local unordered marker orbits, tests full IP and typed Helly, and extends every retained proper residue by all possible outside involutions. Early rejection only uses failure of the necessary irredundance condition. All six rank-four classes and the absence of rank-five residues are outputs of the search; their asserted final counts are checked after enumeration.

```
# Complete finite J1 geometry census by proper-residue extension.
# Relative to the complete TomLib J1 subgroup table, not a geometry census.
LoadPackage("tomlib");;
SizeScreen([100000,24]);;
tom:=TableOfMarks("J1");;G:=UnderlyingGroup(tom);;
ids:=MaximalSubgroupsTom(tom)[1];;
ns:=Length(OrdersTom(tom));; incidence:=SubsTom(tom);;
derived:=Filtered([1..ns-1],i->not ForAny([i+1..ns-1],j->i in incidence[j]));;
if derived<>SortedList(ids) or ns<>40 then Error("maximal subgroup catalogue");fi;
Print("MAXIMAL_CLASSES ",derived," ORDERS
",List(ids,i->Size(RepresentativeTom(tom,i))),"\n");;
CheckCensus:=function(t)
 local n,I,i,j,k,K,L,ss,subs,P,rows,J,common,T;
 n:=Length(t); I:=[1..n];
 # Irredundance is necessary for IP and rejects most global extensions.
 for i in I do
  K:=Group(t{Difference(I,[i])});
  if t[i] in K then return fail;fi;
 od;
 ss:=Combinations(I);
 subs:=List(ss,J->Group(Concatenation([()],t{J})));
 for i in [1..Length(ss)] do for j in [i+1..Length(ss)] do
  if Intersection(subs[i],subs[j])<>subs[Position(ss,Intersection(ss[i],ss[j]))]
   then return fail;fi;
 od;od;
 P:=List(I,i->Group(t{Difference(I,[i])}));
 for i in I do
  rows:=[];
  for j in Difference(I,[i]) do
   K:=Intersection(P[i],P[j]);
   rows[j]:=Set(List(RightTransversal(P[j],K),x->RightCoset(P[i],x)));
  od;
  for J in Combinations(Difference(I,[i])) do
   if Length(J)<2 then continue;fi;
   T:=Intersection(P{J});
   if Length(Intersection(rows{J}))<>Index(T,Intersection(T,P[i]))
    then return fail;fi;
  od;
 od;
 return rec(marking:=t,order:=Size(Group(t)),parabolics:=List(P,Size));
end;;
localReps:=rec();;
```

```
for k in [3,4] do
 localReps.(String(k)):=[];
 for id in ids do
  M:=RepresentativeTom(tom,id);
  iv:=Filtered(Elements(M),x->Order(x)=2);
  A:=Image(ActionHomomorphism(M,iv,OnPoints));
  oo:=OrbitsDomain(A,Combinations([1..Length(iv)],k),OnSets);
  count:=0;
  for o in oo do
   t:=iv{o[1]}; r:=CheckCensus(t);
   if r<>fail then Add(localReps.(String(k)),t);count:=count+1;fi;
  od;
  Print("LOCAL ",k," MAXIMAL ",id," ORDER ",Size(M)," ORBITS ",Length(oo)," REGULAR
  ",count,"\n");
 od;
 Print("LOCAL_TOTAL ",k," ",Length(localReps.(String(k))),"\n");
od;
Print("LOCAL_ENUMERATION_DONE\n");
involClasses:=Filtered(ConjugacyClasses(G),c->Order(Representative(c))=2);;
if Length(involClasses)<>1 then Error("single involution class required");fi;
ivG:=SortedList(Elements(involClasses[1]));;
if Length(ivG)<>1463 then Error("involution count");fi;
a0:=ivG[1];; C0:=Elements(Centralizer(G,a0));;
if Length(C0)<>120 then Error("involution centralizer");fi;
conjCache:=[];;
CanonicalCensus:=function(t)
 local ans,x,p,g,c,key;
 ans:=fail;
 for x in t do
  p:=PositionSorted(ivG,x);
  if not IsBound(conjCache[p]) then
   g:=RepresentativeAction(G,x,a0);
   if g=fail or x^g<>a0 then Error("involution conjugator");fi;
   conjCache[p]:=g;
  fi;
  g:=conjCache[p];
  for c in C0 do
   key:=SortedList(List(t,y->PositionSorted(ivG,y^(g*c))));
   if ans=fail or key<ans then ans:=key;fi;
  od;
 od;
 return ans;
end;;
globalClasses:=[];;fiveClasses:=[];;classParabolics:=[];;
for k in [3,4] do
 keys:=[];unique:=[];
 for t in localReps.(String(k)) do
  key:=CanonicalCensus(t);
  if not key in keys then Add(keys,key);Add(unique,t);fi;
 od;
 Print("GLOBAL_LOCAL_ORBITS ",k," ",Length(unique),"\n");
 tested:=0;passed:=0;start:=Runtime();
 for i in [1..Length(unique)] do
  t:=unique[i];K:=Group(t);
  for x in ivG do
```

```
   if x in K then continue;fi;
   tested:=tested+1;
   r:=CheckCensus(Concatenation(t,[x]));
   if r=fail then continue;fi;
   if k=3 and r.order<>175560 then continue;fi;
   passed:=passed+1;
   key:=CanonicalCensus(r.marking);
   if k=3 and not key in globalClasses then
    Add(globalClasses,key);Add(classParabolics,SortedList(r.parabolics));
    Print("NEW_RANK4 ",key," PARABOLICS ",SortedList(r.parabolics),"\n");
   elif k=4 and not key in fiveClasses then
    Add(fiveClasses,key);
    Print("NEW_RANK5 ",key," GROUP_ORDER ",r.order," PARABOLICS ",r.parabolics,"\n");
   fi;
  od;
  Print("EXTENSION ",k+1," LOCAL ",i,"/",Length(unique)," TESTED ",tested,
   " PASSED ",passed," SECONDS ",(Runtime()-start)/1000,"\n");
 od;
 Print("STAGE_TOTAL ",k+1," TESTED ",tested," ACCEPTED ",passed,"\n");
od;
Print("RANK4_CLASSES ",Length(globalClasses)," RANK5_CLASSES ",Length(fiveClasses),"\n");
if Length(globalClasses)<>6 or Length(fiveClasses)<>0 then Error("geometry classification");fi;
expected:=[[20,20,60,120],[20,20,120,660],[20,60,60,120],
 [20,60,120,120],[60,60,120,120],[20,120,120,660]];;
if SortedList(classParabolics)<>SortedList(expected) then Error("six distinct types");fi;
PrintTo("phase40_compress_operations_exhaustive_classes.json",
 [globalClasses,fiveClasses,classParabolics],"\n");;
Print("COMPRESS_C_EXHAUSTIVE_PASS\n");
QUIT;
```

An explicit lattice bridge for the cube-input halving family

Private additive proof, 10 October 2026. This note does not change the frozen Compress_C files. It addresses precisely ledger N12 and the p=4 identification in N8. Its proof replaces their retained toroidal-identification dependency; independent review of this replacement is still required. The general halving theorem and its regularity proof remain those of Compress_C §§3–4. No additional source is admitted and no compression factor is asserted.

1. Coordinate conventions and the actual face poset

Let r≥2 and let L be the r-dimensional cross-polytope with vertices labelled +e_i,−e_i, 1≤i≤r. A proper face of L chooses at most one vertex from each antipodal pair. Write Q_r for the cubical tessellation of R^r whose vertices are Z^r and whose cells are products of unit intervals and integer singletons. The quotient Q_r/4Z^r has a formal least and greatest face adjoined when viewed as an abstract polytope. These formal faces are also understood below.

For each pair of binary coordinates belonging to +e_i,−e_i, use the Gray map

`(0,0) ↦ 0,  (1,0) ↦ 1,  (1,1) ↦ 2,  (0,1) ↦ 3  in Z/4Z.  (T1)`

Lemma T1. The product of these maps is a rank-preserving face-poset isomorphism 2^L ≅ Q_r/4Z^r. Binary weight modulo two becomes $\sum$_i x_i modulo two in the cubical coordinates.

Proof. In Danzer's construction a proper face of 2^L allows the bits indexed by a face of L to vary, and fixes every other bit. Thus in any antipodal pair either both bits are fixed or exactly one bit varies. Under (T1), varying a single bit describes precisely one edge of the four-cycle: its four possible edges arise from the two choices of varying bit and the two values of the fixed bit. A face therefore becomes a product of four-cycle vertices and edges. Conversely every such product uniquely specifies the varying bit in each free pair and the remaining fixed bits. Its dimension is the number

of free pairs. Inclusion is preserved and reflected coordinatewise, and formal faces match. These products are exactly the cubical cells modulo 4Z^r, with their incidence. The four cases of (T1) have binary parities 0,1,0,1 respectively, proving the last assertion. ■

This is a face-poset identification, rather than an inference from orders. It also handles the dual in the target source explicitly. For a cubical face F with free coordinate set S, write its free intervals as [m_i,m_i+1] for i∈S and its fixed coordinates as m_j for j∉S. The dual face F* has fixed coordinate m_i+1/2 for i∈S and free interval [m_j−1/2,m_j+1/2] for j∉S. This reverses containment: whenever an endpoint is fixed in a subface of F, the corresponding free interval of the dual contains the fixed midpoint of F*. It is a bijection onto the shifted cubical tessellation, with dimensions adding to r. Subtracting (1/2,…,1/2) identifies that tessellation with Q_r. The construction commutes with translations by 4Z^r, and consequently induces an order-reversing bijection on the quotient. Thus for the r-cube K=L*, the source's polytope (2^{K*})* is indeed the cubical toroid {4,3^(r−2),4}_(4,0,…,0). The string is {4,4} when r=2. Reversing the source marking moves its last-end halving to first-end halving of 2^L. Type reversal is kept explicit; no unproved compatibility of a chosen self-duality with halving is needed.

2. The marking and the exact affine quotient

Use the following reflections of R^r:

```
a(x_1,x_2,…,x_r) = (1−x_1,x_2,…,x_r),
b_i swaps x_i and x_(i+1)                          (1≤i<r),
c(x_1,…,x_(r−1),x_r) = (x_1,…,x_(r−1),−x_r).       (T2)
```

They generate W=Z^r⋊B_r, where B_r is the signed permutation group. Indeed the b_i,c generate B_r, including the reflection x_1↦−x_1; its product with a is translation by ±e_1. Signed permutations then supply every integer translation. The simplex 1/2≥x_1≥x_2≥⋯≥x_r≥0 is a fundamental chamber. One way to verify this without a presentation theorem is to translate each coordinate to [−1/2,1/2], make it nonnegative, and order the coordinates; a point with strict inequalities has a unique such representative. Its reflected walls are the displayed generators. Its vertices are the barycentres of a nested flag of cubical faces, so its chamber system is the barycentric subdivision of Q_r.

The marking in (T2) also agrees with Danzer's marking under (T1). Flipping the +e_1 bit induces x_1↦1−x_1 modulo four; interchanging antipodal pairs induces the b_i; swapping the two bits in the last pair induces x_r↦−x_r. These are the usual base vertex/base flag generators of 2^L. Hence the isomorphism of Lemma T1 identifies the marked action, not only the unmarked face poset. The action of W on Q_r/4Z^r has kernel exactly Λ=4Z^r: an affine signed permutation acting trivially on all vertices has translation zero modulo four, and its linear part is the identity modulo four; testing the unit vectors forces that signed permutation to be the identity.

Put b=b_1 and replace a by

```
d=aba: (x_1,x_2,x_3,…,x_r)
       ↦(1−x_2,1−x_1,x_3,…,x_r).              (T3)
```

Let E={v∈Z^r : ∑v_i is even}.

Lemma T2. H=⟨d,b_1,…,b_(r−1),c⟩ is exactly E⋊B_r. It has index two in W and contains Λ. Its fundamental chamber is

```
x_1≥x_2≥⋯≥x_r≥0,   x_1+x_2≤1.                 (T4)
```

Its chamber system is the halved cubical geometry, and the marked finite halving group and geometry are H/Λ and the quotient of that chamber system by Λ, respectively.

Proof. Every displayed generator preserves the parity of an integral vertex's coordinate sum, so H≤E⋊B_r. It contains B_r. The product of d with the B_r-reflection (x_1,x_2)↦(−x_2,−x_1) is translation by e_1+e_2. Conjugating by B_r gives every ±e_i±e_j. These span E: subtract multiples of e_i−e_r, i<r, to leave an even multiple of e_r, and use 2e_r=(e_r+e_1)+(e_r−e_1). This works also for r=2. Thus equality holds and the index is two. Since every 4e_i is in E, Λ≤H.

The region (T4) is the union of the preceding W chamber and its reflection in the a-wall x_1=1/2. On their common wall the two chambers glue; the other wall x_1=x_2 is paired with its a-image x_1+x_2=1. All other walls agree.

Because W=H⊔Ha and the interiors of W chambers are disjoint, these doubled chambers tile R^r under H, each once. Their wall reflections are exactly d,b_1,…,b_(r−1),c. Equivalently, their chamber labels are the old cubical flags with the initial vertex chosen in one parity class, pairing the two old flags differing only at that vertex. The two ends of the old edge become the two new vertex types; the higher faces retain their type. This is precisely the graph-halving construction of Compress_C §3.

Finally, translations in Λ preserve both vertex colors and take cubical faces to cubical faces. The same flag pairing therefore commutes with Λ. Every finite chamber has a lift, and two lifts yield the same finite chamber exactly when they differ by Λ. Wall adjacencies are the marked reflections modulo Λ. The maximal parabolics are the images of the corresponding wall subgroups, so the typed coset geometry also agrees. Regularity, thinness and residual connectedness of the finite geometry follow from the already proved halving theorem; they are not inferred solely from being a group quotient. ■

For r≥3, the walls (T4) have the affine fork diagram B̃_r: d and b_1 commute; both meet b_2 with product order three, successive b_i have order three, and b_(r−1)c has order four. All other distinct wall pairs commute. This follows directly from (T2)–(T3), or from their mirror angles. Thus its rank-(r+1) quotient has the source's type {4,3^(r−3),3/3}. Here the quotient lattice is exactly 4Z^r in the original coordinates; the subscript (4,0,…,0) denotes the lattice generated by that vector's signed-permutation orbit. This proves the source's all-rank toroidal identification, including the marking. Its group order is [E:4Z^r]|B_r|=$(4^{r/2)2}$r r!=2^(3r−1)r!, agreeing with the separate parity-module calculation. The order is a check, not the proof.

The coordinate convention can be checked directly against Ens's primary paper: its equations (3.1), (4.1) are (T2), (T3) for r=3; its §2 definition of Λ_s uses the signed-permutation orbit of s, and Theorem 4.3 denotes the corresponding Coxeter-complex quotient by that lattice. Thus s=(4,0,0) here is literally the same parameter. The target extends this convention to larger r. No classification theorem about all invariant lattices is needed. Ens, *Rank 4 toroidal hypertopes*, pp.70–75.

3. The rank-three square map and its changed lattice basis

When r=2, reorder the halved types as (even vertex, old square, odd vertex). An old unit square joins its two even corners as an edge of the new map; the four even neighbors of an old odd vertex bound a new square face. This describes the ordinary square tessellation with vertex lattice E and edge directions

`f_1=(1,1),  f_2=(−1,1).` (T5)

Every even vertex has four such incident edges, every displayed quadrilateral is a face, and the incidences are exactly those of halving. Thus this is an isomorphism of maps, not a comparison of their counts. In this basis

`4e_1=2f_1−2f_2,  4e_2=2f_1+2f_2.`

The quotient lattice has basis (2,−2),(2,2) in square coordinates. Negating the first vector gives the customary pair (−2,2),(2,2), so the map is {4,4}_(2,2). Lemma T1 and the explicit duality already identify the original map as {4,4}_(4,0). This completes both p=4 names in ledger N8.

4. Source and proof accounting

The frozen target is Montero–Weiss, arXiv:2001.11082v1, §§3–4, especially pp.9–10. The cube construction and the precise target names were reread from the existing bound primary text. Ens's primary paper was selectively read in the web-rendered complete article at the link above: pp.70–75, definitions of the lattice notation, equations (3.1), (4.1), the translation description, and Theorem 4.3. Its entire classification proof and later sections were not independently reviewed. Publisher access failed; no local PDF bytes or page screenshots of Ens are claimed. Its definitions are used only to match notation; the particular quotient is proved above.

New dependency delta: remove the retained toroidal identifications from C ledger N12 and the p=4 part of N8 **if this note passes independent review**. Retain Danzer's general construction/full-group theorem where C uses it, the reviewed geometric halving theorem, and C's other explicit background. The argument here proves this cross-polytope input directly and needs no new exhaustive computation, numerical experiment, catalogue, or Lean claim. It does not extend C's intake to Ens's classification paper, nor replace any of that paper's classification or chirality results.